OXFORD

Journal of AOAC INTERNATIONAL, 2026, **109(4),** 511–525

https://doi.org/10.1093/jaoacint/qsag007

Advance Access Publication Date: 3 February 2026

Research Article

Food Chemical Contaminants

Validation of the U.S. Food and Drug Administration's Method for Detection of Per- and Polyfluoroalkyl Substances Analytes in Food and Feed: First Action 2025.06

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Abstract

Background: The AOAC Stakeholder Panel on Strategic Food Analytical Methods approved AOAC INTERNATIONAL *Standard Method Performance Requirement* (SMPR®) 2023.003 for per- and polyfluoroalkyl substances (PFAS) in produce, beverages, dairy products, eggs, seafood, meat products, and feed. The U.S. Food and Drug Administration's (FDA's) method is a single-laboratory validated (SLV) method for determination of 30 PFAS in food and feed.

Objective: In response to a call for minimum method performance characteristics and analytical requirements, the FDA's method for detection of PFAS in food and feed was validated in a single-laboratory study with comparison to SMPR 2023.003.

Methods: Samples were extracted using a Quick, Easy, Cheap, Effective, Rugged, Safe (QuEChERS) method followed by solid-phase extraction (SPE) cleanup and concentration using nitrogen. Analysis was performed using isotope dilution liquid chromatography–tandem mass spectrometry (LC–MS/MS). The method was validated for parameters such as LOQ, recovery, repeatability, and system suitability, and reference materials were analyzed.

Results: A full validation was conducted in European Union (EU)-regulated matrixes (eggs, shrimp, clams, salmon, chicken, and beef liver) and other matrixes (produce, milk, baby food, ground coffee, fish oil, protein powder, and corn snaplage) with satisfactory performances both in terms of recovery and reproducibility. Recovery percentages at target LOQs and two additional levels for perfluorooctane sulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), and perfluorohexane sulfonic acid (PFHxS) were between 80 and 120% and repeatability (RSD_r) results were ≤20% for EU-regulated matrixes. For other matrixes recovery percentages were between 65 and 135% and repeatability results were ≤25%. All were within acceptable ranges as per SMPR 2023.003.

Conclusion: The FDA's method for detection of 30 PFAS in food and feed compares favorably with the requirements of SMPR 2023.003 and should be adopted as a First Action method.

Highlights: The FDA's single-laboratory validated method for PFAS detection in food and feed meets the requirements of SMPR 2023.003.

Per- and polyfluoroalkyl substances (PFAS) are used in a variety of domestic and industrial applications due to their surface tension-lowering properties and strong chemical stability (1). They have been used in electroplating, textile stains, soil repellents, non-stick cookware, cosmetics, and fluoropolymer production (2). Widespread production, application, and disposal of PFAS-containing products have led to their detection in the environment, humans, and wildlife (1). Certain PFAS are identified as toxic and bio-accumulative pollutants, and there is increasing awareness about their potential environmental and toxicological impacts (3). This has led to advances in analytical methodology to develop and optimize analytical methods to monitor PFAS in the environment and humans at trace levels (4, 5).

Humans can be exposed to PFAS either through dust inhalation, contaminated food, and water, or through environmental sources (6). The U.S. Food and Drug Administration (FDA) has been monitoring samples from the FDA's total diet study program for PFAS contamination since 2018. The FDA has developed analytical methods for detection of PFAS in a wide variety of food and feed with the first method published on the FDA website in 2019 (C-010.01; 7, 8). This method was modified and expanded to include additional food matrixes and published in 2021 (C-010.02; 9). In 2022, the European Union Reference Laboratory (EURL) developed a guidance document on analytical parameters for the determination of selected PFAS in food and feed. This guidance document contained recommendations for validation

Received: 17 November 2025. **Revised:** 09 January 2026. **Accepted:** 18 January 2026

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parameters and analytical method performance criteria for 30 PFAS analytes (5, 9). As a result, the FDA expanded analytical method C-010.02 to include 30 PFAS and performed a single-laboratory validation using seven food and two feed matrixes, published as C-010.03 in 2024 (9). The results were compared to Commission Regulation European Union (EU) 2023/915, which established maximum levels for perfluorooctane sulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), and perfluorohexane sulfonic acid (PFHxS) in meat, edible offal, eggs, fishery products, and bivalve molluscs (9).

The effectiveness of detecting suspected contaminated food or feed is highly dependent on the quality and performance of laboratory methods (10). In December 2023, the AOAC INTERNATIONAL Working Group on Per- and Polyfluoroalkyl Substances developed a *Standard Method Performance Requirement* (SMPR®) for PFAS in foods, setting required LOQs for PFAS analytes (11). SMPR 2023.003 requires a mass spectrometry-based method that quantifies 30 PFAS analytes in 11 targeted matrixes with stricter validation requirements for PFOS, PFOA, PFNA, and PFHxS in European Union (EU) regulated matrixes (5). As a result, the FDA method was evaluated for performance in reaching the LOQ for foods following validation requirements set by AOAC.

AOAC *Official Method*℠
2025.06
Detection of Per- and Polyfluoroalkyl Substances
Analytes in Food and Feed
Liquid Chromatography-Tandem Mass Spectrometry
First Action 2025

[Applicable for determination of 30 per- and polyfluoroalkyl substances (PFAS) analytes (including selected PFAS analytes—perfluorooctane sulfonic acid (PFOS), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), and perfluorohexane sulfonic acid (PFHxS) mentioned in *Standard Method Performance Requirement* (SMPR®) 2023.003 in produce, beverages, dairy products, eggs, seafood, meat products, and feed using LC–MS/MS (11).]

Caution: Use appropriate personal protective equipment such as laboratory coat, safety glasses, and gloves. PFAS chemicals are prevalent in all laboratory environments and special care must be taken to prevent false positives due to accidental and/or routine laboratory contamination. Individuals responsible for sampling should not wear clothing or gloves that contain fluorochemicals. Additional considerations that need to be taken are the use of LC–MS grade solvents and use of a delay column. Prior to sample analysis it is crucial to analyze all solvents and method blanks using LC–MS/MS to make sure there is no background contamination coming from solvents and laboratory equipment. Sources of potential contamination during sample preparation include solvents, syringe filters, centrifuge tubes, solid-phase extraction (SPE) sorbents, septa, and others. Use of a delay column delays elution of PFAS that come from instrument parts, mobile phase lines, and solvents, which would interfere with the quantitation of PFAS in food.

Principle

A modified Quick, Easy, Cheap, Effective, Rugged, Safe (QuEChERS) technique is performed for the extraction of PFAS from foods. All samples are spiked with isotopically labeled standards prior to extraction with acetonitrile and formic acid. After extraction, a dispersive SPE (dSPE) step is performed followed by further cleanup using weak anion exchange (SPE). For matrixes where there is a detection of both PFOS and taurochenodeoxycholic acid (TCDCA), an additional cleanup is performed with the QuECHERS extract using ENVI-Carb cartridges for accurate PFOS quantification. An internal standard is added to the final extract and analyzed using LC–MS/MS. For perfluorobutanoic acid (PFBA) and perfluoropentanoic acid (PFPeA) where two abundant ion transitions are not available using low-resolution mass spectrometry, high-resolution mass spectrometry (HRMS) was used for verifying quantification.

Chemicals and Reagents

Note: The following mentioned commercial sources are for reference purposes only. Equivalent chemicals and reagents may be used but it is recommended to make sure chemicals and reagents are PFAS-free.

- **(a)** *Water.*—LC/MS grade Optima (Cat. No. W6-4; Thermo Fisher Scientific, Waltham, MA, USA) or equivalent.
- **(b)** *Acetonitrile.*—LC/MS grade Optima (Cat. No. A955-4; Thermo Fisher Scientific) or equivalent.
- **(c)** *Formic acid.*—LC/MS grade Optima >99% (Cat. No. A11750; Millipore Sigma, St. Louis, MO, USA) or equivalent.
- **(d)** *Methanol.*—LC/MS Grade (Cat. No. 60-009-61; Honeywell International Inc., Morristown, NJ, USA) or equivalent.
- **(e)** *Acetic acid, ammonium salt.*—98% for analysis (Cat. No. 14267-25G; Acros Organic, Geel, Belgium) or equivalent.
- **(f)** *1-Methyl piperidine.*—99% (Cat. No. L03398; Alfa Aeser, Tewksbury, MA, USA).
- **(g)** *Original QuEChERS extraction salt with 6000 mg $MgSO_4$ and 1500 mg NaCl.*—(Cat. No. ECMSSCFS-MP; (UCT), Bristol, PA, USA) or equivalent.
- **(h)** *QuEChERS dSPE with 900 mg $MgSO_4$, 300 mg PSA, 150 mg graphitized carbon black.*—(Cat No. ECMPSCB-MP; (UCT), Bristol, PA, USA) or equivalent.
- **(i)** *Ammonium hydroxide.*—Certified ACS Plus 14.8 N (Cat. No. A669-500; ThermoFisher Scientific) or equivalent.
- **(j)** *Isotopically labeled PFAS analytical standards custom mix.*—(Cat. No. *FDA12-MPFAS;* Wellington Laboratories, Guelph, ON, Canada).
- **(k)** *Isotopically labeled PFAS analytical standards fluorotelomer sulfonates (FTS) custom mix.*—(Cat. No. *ES-5661;* .Cambridge Isotopes, Tewksbury, MA, USA).
- **(l)** *Native PFAS analytical standards 30 analyte custom mix.*—(Cat. No. 65947; Absolute Standards, Hamden, CT, USA).
- **(m)** *N-ethyl-d_5-perflouro-1-octanesulfonamidoacetic acid.*—(Cat. No. d5-N-EtFOSAA; Wellington Laboratories, Guelph, ON, Canada).

Apparatus

- **(a)** *Digital pulse mixer.*—Capable of 1500 rpm with pulse 70 (Glas-Col, Terre Haute, IN, USA).
- **(b)** *Sorvall Legend XTR centrifuge.*—(Thermo Fisher Scientific, Waltham, MA, USA).
- **(c)** *Nitrogen evaporation system.*—Turbovap LV (Biotage, Uppsala, Sweden)
- **(d)** *Positive-pressure manifold 48 processor.*—(Cat. No. PPM-48; Agilent, Santa Clara, CA, USA).
- **(e)** *Polypropylene (PP) ultra-high performance centrifuge tubes.*—50 mL (VWR Radnor, PA, USA) or equivalent.
- **(f)** *PP conical centrifuge tubes.*—15 mL Falcon (Thermo Fisher Scientific) or equivalent.

(g) *Round-bottom PP test tubes with cap.*—14 mL Falcon (Cat. No. 352059; Millipore Sigma) or equivalent.

(h) *PP screw top vials.*—300 µL PP screw-top vials for calibration standards (MicroSolv, Wilmington, NC, USA).

(i) *Screw caps, silicone rubber/PP septa.*—Caps for calibration standards (Microsolv).

(j) *PP screw-top vials.*—2 mL vials for standard storage (MicroSolv).

(k) *Screw caps, solid, silicone rubber/PP liners.*—Caps for standard storage (MicroSolv).

(l) *Acrodisc nylon syringe filters.*—0.2 µm (Pall Corp., Port Washington, NY, USA).

(m) *PP/PE luer lock syringes, 5 mL.*— (Millipore Sigma).

(n) *Nano filter vial with caps for PFAS use.*—0.2 µm nylon (Cat No. 14638; Thermo Fisher Scientific).

(o) *SPE cartridge.*—Strata^{TM}-XL-AW 100 µm Polymeric Weak Anion 200 mg/6 mL tubes (Phenomenex, Torrance, CA, USA).

(p) *ENVI-Carb cartridge.*—SupelClean ENVI-Carb, 250 mg/6 mL (Millipore Sigma).

(q) *Cryo-Blade cryogenic grinder.*—(Cat No. CG-900, 115 VAC, 20 A; Cole-Palmer, Vernon Hills, USA).

(r) *Liquid chromatograph and columns.*—Nexera X2 HPLC (Shimadzu Corp., Kyoto, Japan) with binary pump, degasser, autosampler, and column with thermostat compartment. All compounds are separated using a Waters XBridge C18, 3.5 µm, 150 mm × 2.1 mm analytical column (Waters Corp., Milford, MA, USA), at a constant flow rate of 0.3 mL/min and a column temperature of 40°C. The guard column is a Waters Vanguard Acquity BEH C18, 1.7 µm, 2.1 mm × 5 mm (Waters Corp., Milford, MA, USA), and the delay column is a Waters XBridge C18 5 µm, 2.1 mm × 50 mm, (Waters Corp.). The operating conditions are as follows. An injection volume of 3 µL is used. Mobile phase A is 5 mM ammonium acetate and 5 mM 1-methyl piperidine (1-MP) in water and mobile phase B is 100% methanol. The addition of 1-MP reduces background and enhances ionization in negative mode resulting in lower method detection limits. The gradient starts at 10% mobile phase B, remains constant for 3 min, followed by an increase to 40% mobile phase B in 0.1 min and then to 90% mobile phase B in the next 23.60 min. The column is equilibrated back to the initial conditions of 10% mobile phase B in 0.1 min and maintained at the same conditions for approximately 9 min.

(s) *Triple quadrupole mass spectrometer.*—SCIEX 6500^{+} QTRAP hybrid triple quadrupole/linear ion trap mass spectrometer with an electrospray (ESI) ion source (SCIEX, Framingham, MA, USA). All data is collected in negative multiple reaction monitoring (MRM) mode with the following ESI parameters: curtain gas set to 40 au, CAD gas set to medium, ion spray voltage of −4500 V, source temperature of 350°C, and gas 1 and gas 2 set to 50 au. MS/MS conditions for each ion transition are mentioned in Table 1. A typical chromatogram is given in Figure 1.

(t) *High-resolution mass spectrometer.*—Q-Exactive Orbitrap mass spectrometer (Thermo Fisher Scientific). The Orbitrap mass spectrometer is operated in negative ion polarity with a full scan method in the mass range of *m/z* 100–1200, resolving power of 70k, Automatic Gain Control (AGC) target of 1e^{6}, and a maximum injection time of 250 ms. In-source fragmentation is minimized by tuning heated electrospray ionization (HESI) source parameters for PFBA and PFPeA; the optimized parameters are sheath gas flow rate 35 au, auxiliary gas flow rate 10 au, spray voltage −2500 V, capillary temperature 350°C, S-lens RF level 25, and auxiliary gas heater temperature 310°C.

General Preparation

(a) *Preparation of native PFAS stock solutions.*—These stock solutions are used for calibration curve preparation and validation spikes.

(1) *1000 ng/mL native PFAS stock solution.*—Add 2 mL 2000 ng/mL PFAS analytical standard custom mix from Absolute Standards (30 native compounds in Table 2) to 2 mL methanol.

(2) *100 ng/mL native PFAS stock solution.*—Add 1 mL 1000 ng/mL stock solution to 9 mL methanol.

(3) *10 ng/mL native PFAS stock solution.*—Add 1 mL 100 ng/mL stock solution to 9 mL methanol.

(b) *Preparation of 200 ng/mL isotopically labeled PFAS surrogate stock solution* (SS).—To prepare 200 ng/mL isotopically labeled PFAS surrogate stock solution, combine 2 mL 1000 ng/mL Cambridge isotope mass-labeled PFAS mixture, containing four FTSs, and 1 mL 2000 ng/mL Wellington mass-labeled PFAS mixture, containing FDA12-MPFAS, with 7 mL methanol. This solution is used during sample extractions and calibration curve preparation.

(c) *Preparation of 200 ng/mL isotopically labeled internal standard solution (IS).*—To prepare 200 ng/mL N-ethyl-d_5-perflouro-1-octanesulfonamidoacetic acid (d_5-N-EtFOSAA), add 0.04 mL 50 µg/mL d_5-N-EtFOSAA to 9.96 mL methanol.

(d) *Preparation of calibration standards.*—Calibration standards are prepared from native PFAS stock solutions according to Table 3.

(e) *Preparation of continuing calibration verification (CCV) standard.*—A duplicate solution of 1 ng/mL calibration standard is prepared as mentioned in Table 3 above and used as the CCV standard.

(f) *Preparation of mobile phases A and B.*—

(1) *Preparation of mobile phase A (5 mM ammonium acetate in water and 5 mM 1-MP).*—Weigh 0.38 ± 0.01 g ammonium acetate and add it to 1000 mL LC/MS Optima water. Then add 0.5 mL 1-MP to this and mix well. Use an amber-colored solvent bottle to prepare mobile phase A.

(2) *Preparation of mobile phase B.*—100% LC–MS grade methanol.

(g) *Preparation of SPE solution for cleanup (0.3%, w/v, ammonium hydroxide in methanol).*—Add 6 mL 14.8 N ammonium hydroxide solution to a 1000 mL volumetric flask and make up to volume with methanol.

Sample Preparation

(a) *Sample homogenization.*—Homogenize chicken, shrimp, salmon, clams, beef liver, and pet food using an IKA tube mill (IKA Works Inc, Wilmington, NC, USA) at a rate of 5000 rpm for approximately 2 min. Homogenize fibrous matrixes such as snaplage using a Cole-Parmer CG-900 Cryo-Blade cryogenic grinder.

(b) *Sample extraction.*—Three separate sample extraction protocols are used based on the sample matrix (food or feed)

Table 1. MS/MS conditions for the monitored transitions on a 6500 plus QTRAP

ID	Q1 mass (*m/z*)	Q3 mass (*m/z*)	DP, V	EP, V	CE, V	CXP, V
Internal standard						
d_5-N-EtFOSAA[a]	589	419	−50	−10	−30	−20
d_5-N-EtFOSAA	589	219	−50	−10	−38	−20
Surrogates						
M3PFBA[a]	216	172	−17	−8	−12	−14
M3PFPeA[a]	266	222	−17	−6	−11	−28
M5PFHxA[a]	318	273	−13	−10	−14	−12
M8PFOA[a]	421	376	−36	−8	−13	−20
M8PFOA	421	172	−19	−5	−25	−7
MPFUdA[a]	565	520	−32	−4	−17	−45
MPFUdA	565	269	−35	−5	−27	−22
MPFDoA[a]	615	570	−48	−3	−19	−45
MPFDoA	615	169	−51	−5	−38	−17
M2PFTeDA[a]	715	670	−38	−3	−23	−52
M2PFTeDA	715	169	−36	−3	−38	−18
M3PFBS[a]	302	80	−88	−6	−73	−9
M3PFBS	302	99	−85	−6	−36	−8
M3PFHxS[a]	402	80	−60	−10	−81	−15
M8PFOS[a]	507	80	−100	−5	−125	−15
M8PFOS	507	99	−100	−5	−100	−15
M3HFPO[a]	287	169	−19	−12	−10	−25
M3HFPO	287	185	−41	−5	−35	−35
M8FOSA[a]	506	78	−101	−9	−100	−8
M8FOSA	506	64	−120	−9	−165	−28
$^{13}C_2$, D_4 4:2 FTS[a]	333	312	−60	−10	−30	−29
$^{13}C_2$, D_4 4:2 FTS	333	82	−60	−10	−63	−12
$^{13}C_2$, D_4 6:2 FTS[a]	433	412	−80	−10	−33	−13
$^{13}C_2$, D_4 6:2 FTS	433	82	−80	−10	−67	−13
$^{13}C_2$, D_4 8:2 FTS[a]	533	512	−100	−10	−49	−23
$^{13}C_2$, D_4 8:2 FTS	533	82	−100	−10	−81	−34
$^{13}C_2$, D_4 10:2 FTS[a]	633	612	−100	−10	−45	−19
$^{13}C_2$, D_4 10:2 FTS	633	82	−100	−10	−85	−10
Natives						
PFBA[a]	213	169	−10	−6	−13	−19
PFPeA[a]	263	219	−20	−8	−11	−20
PFHxA[a]	313	269	−10	−12	−13	−45
PFHxA	313	119	−23	−12	−27	−14
PFHpA[a]	363	319	−17	−10	−14	−26
PFHpA	363	169	−32	−11	−25	−24
PFOA[a]	413	369	−43	−7	−16	−25
PFOA	413	219	−24	−5	−23	−25
PFNA[a]	463	419	−38	−11	−15	−37
PFNA	463	269	−40	−5	−24	−13
PFDA[a]	513	469	−15	−10	−16	−29
PFDA	513	269	−20	−10	−26	−17
PFUdA[a]	563	519	−38	−9	−18	−45
PFUdA	563	269	−39	−7	−27	−25
PFDoA[a]	613	569	−33	−7	−19	−52
PFDoA	613	169	−33	−8	−39	−17
PFTrDA[a]	663	619	−39	−7	−21	−18
PFTrDA	663	169	−39	−7	−39	−18
PFTeDA[a]	713	669	−42	−12	−25	−51
PFTeDA	713	169	−49	−6	−40	−19
PFBS[a]	299	80	−44	−10	−70	−11
PFBS	299	99	−35	−4	−36	−15
PFPeS[a]	349	99	−80	−9	−80	−12
PFPeS	349	119	−53	−10	−40	−18
PFHxS[a]	399	99	−108	−6	−84	−8
PFHxS	399	169	−66	−5	−42	−20
PFHpS[a]	449	99	−58	−8	−84	−24
PFHpS	449	169	−68	−8	−41	−27
PFOS[a]	499	80	−150	−4	−120	−10
PFOS	499	99	−150	−4	−100	−10
PFNS[a]	549	80	−72	−10	−142	−8
PFNS	549	99	−110	−10	−124	−12
PFDS[a]	599	80	−136	−8	−171	−14
PFDS	599	99	−139	−7	−141	−14
PFUdS[a]	649	80	−180	−4	−160	−12
PFUdS	649	99	−170	−4	−140	−11
PFDoS[a]	699	80	−79	−11	−171	−9

(continued)

Table 1. (continued)

	Internal standard					
ID	**Q1 mass (*m/z*)**	**Q3 mass (*m/z*)**	**DP, V**	**EP, V**	**CE, V**	**CXP, V**
PFDoS	699	99	−130	−10	−158	−10
PFTrDS[a]	749	80	−76	−11	−177	−37
PFTrDS	749	99	−79	−13	−174	−40
FOSA[a]	498	78	−138	−11	−93	−7
FOSA	498	478	−130	−13	−41	−36
DONA[a]	377	251	−25	−8	−15	−20
DONA	377	85	−20	−7	−39	−10
HFPO−DA[a]	285	169	−78	−6	−11	−27
HFPO−DA	285	185	−20	−5	−21	−27
9Cl−PF3ONS[a]	531	351	−100	−14	−38	−13
9Cl−PF3ONS	531	83	−100	−4	−92	−9
11Cl−PF3OUdS[a]	631	451	−34	−9	−40	−12
11Cl−PF3OUdS	631	199	−20	−10	−36	−11
4:2 FTS[a]	327	307	−50	−6	−28	−24
4:2 FTS	327	81	−40	−11	−62	−10
6:2 FTS[a]	427	407	−31	−11	−34	−38
6:2 FTS	427	81	−41	−12	−88	−12
8:2 FTS[a]	527	507	−69	−11	−37	−29
8:2 FTS	527	81	−70	−11	−109	−36
10:2 FTS[a]	627	607	−54	−6	−46	−48
10:2 FTS	627	81	−68	−6	−129	−15
Cholic acid transitions monitored for potential interferences (covers TDCA, TUDCA, and TCDCA)						
Cholic acid 1	498	124	−161	−14	−62	−16
Cholic acid 2	498	80	−107	−11	−152	−9

[a] Primary MRM transition used for quantification.

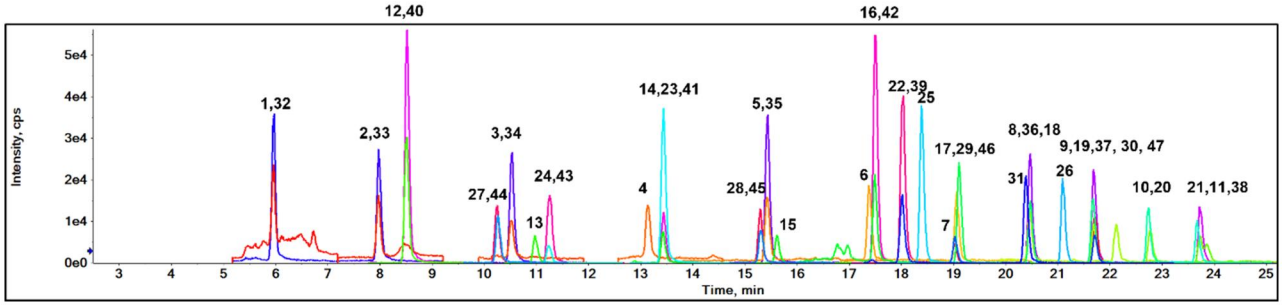


Figure 1. Typical LC–MS/MS chromatogram of PFAS analytes. 1. PFBA, 2. PFPeA, 3. PFHxA, 4. PFHpA, 5. PFOA, 6. PFNA, 7. PFDA, 8. PFUnA, 9. PFDoA, 10. PFTrDA, 11. PFTeDA, 12. PFBS, 13. PFPeS, 14. PFHxS, 15. PFHpS, 16. PFOS, 17. PFNS, 18. PFDS, 19. PFUnDS, 20. PFDoS, 21. PFTrDS, 22. FOSA, 23. DONA, 24. HFPO-DA, 25. 9Cl-PF3ONS, 26. 11Cl-PF3OUdS, 27. 4:2 FTS, 28. 6:2 FTS, 29. 8:2 FTS, 30. 10:2 FTS, 31. d5 N-EtFOSAA, 32. M3PFBA, 33. M3PFPeA, 34. M5PFHxA, 35. M8PFOA, 36. MPFUdA, 37. MPFDoA, 38. M2PFTeDA, 39. M8 FOSA, 40. M3PFBS, 41. M3PFHxS, 42. M8PFOS, 43. M3HFPO-DA, 44. $_{13}C_2D_4$ 4:2 FTS, 45. $_{13}C_2D_4$ 6:2 FTS, 46. $_{13}C_2D_4$ 8:2 FTS, 47. $_{13}C_2D_4$ 10:2 FTS.

and sample size. Sample matrixes and sample sizes used are listed in Table 4.—

(1) *Sample preparation for food (matrixes such as chicken, shrimp, salmon, clams, beef liver, pet food, brewed coffee).*—
   (*a*) Add 5 g of food sample to a 50 mL PP centrifuge tube. Weigh additional 5 g food samples as needed for duplicate or triplicate analysis. Weigh 5 mL water for method blank analysis.
   (*b*) Add 50 µL 200 ng/mL isotopically labeled SS to give a final concentration of 1 ng/mL in the final extract. Add 5 mL water, 10 mL acetonitrile, and 150 µL formic acid to the 50 mL PP conical centrifuge tube and shake it vigorously for 1 min.
   (*c*) Add the contents of a QuEChERS salt packet (original extraction salt ECMSSCFS-MP from UCT with 6000 mg $MgSO_4$ and 1500 mg NaCl). If clumping occurs, shake the sample by hand or vortex mix until it is homogenous. Place on a Glas-Col shaker at 1500 rpm with pulse set to 70 for 5 min.
   (*d*) Centrifuge for 5 min at 10000 g. Add the supernatant to a 15 mL PP conical centrifuge tube with dSPE sorbent (ECMPSCB-MP from UCT with 900 mg $MgSO_4$, 300 mg PSA, 150 mg graphitized carbon black). Vortex mix or shake for 2 min and centrifuge for 5 min at 10000 g.
   (*e*) Filter the supernatant with a 0.2 µm nylon syringe filter and transfer to a 15 mL conical centrifuge tube.
   (*f*) For SPE cleanup, take 1 mL of the filtered supernatant and transfer to a 15 mL centrifuge tube. Then add 11 mL water.

(2) *Sample preparation for feed and dried foods (snaplage and protein powder).*—
   (*a*) Add 1 g of dry food/feed sample (snaplage or protein powder) to a 50 mL PP centrifuge tube. Weigh additional 1 g samples as needed for duplicate or triplicate analysis and add 1 g water to a centrifuge tube for method blank analysis.
   (*b*) Add 10 µL 200 ng/mL isotopically labeled SS for a final concentration of 1 ng/mL in the final extract.

Table 2. Names of PFAS native, surrogate, and internal standard compounds

Native PFAS				
Acronym	**Name**	**CAS**	**Formula**	**MW**
PFBA	Perfluorobutanoic acid	375–22-4	$C_4F_7O_2$	214
PFPeA	Perfluoropentanoic acid	2706–90-3	$C_5HF_9O_2$	264
PFHxA	Perfluorohexanoic acid	307–24-4	$C_6HF_{11}O_2$	314
PFHpA	Perfluoroheptanoic acid	375–85-9	$C_7HF_{13}O_2$	364
PFOA	Perfluorooctanoic acid	335–67-1	$C_8HF_{15}O_2$	414
PFNA	Perfluorononanoic acid	375–95-1	$C_9HF_{17}O_2$	464
PFDA	Perfluorodecanoic acid	335–76-2	$C_{10}HF_{19}O_2$	514
PFUnA	Perfluoroundecanoic acid	2058–94-8	$C_{11}HF_{21}O_2$	564
PFDoA	Perfluorododecanoic acid	307–55-1	$C_{12}HF_{23}O_2$	614
PFTrDA	Perfluorotridecanoic acid	72629–94-8	$C_{13}HF_{25}O_2$	664
PFTeDA	Perfluorotetradecanoic acid	376–06-7	$C_{14}HF_{27}O_2$	714
PFBS	Perfluorobutanesulfonic acid	375–73-5	$C_4HF_9O_3S$	300
PFPeS	Perfluoropentanesulfonic acid	2706–91-4	$C_5HF_{11}O_3S$	350
PFHxS	Perfluorohexanesulfonic acid	355–46-4	$C_6HF_{13}O_3S$	400
PFHpS	Perfluoroheptanesulfonic acid	375–92-8	$C_7HF_{15}O_3S$	450
PFOS	Perfluorooctanesulfonic acid	1763–23-1	$C_8HF_{17}O_3S$	500
PFNS	Perfluorononanesulfonic acid	68259–12-1	$C_9HF_{19}O_3S$	550
PFDS	Perfluorodecanesulfonic acid	335–77-3	$C_{10}HF_{21}O_3S$	600
PFUnDS	Perfluoroundecanesulfonic acid	749786–16-1	$C_{11}HF_{23}O_3S$	650
PFDoS	Perfluorododecanesulfonic acid	79780–39-5	$C_{12}HF_{25}O_3S$	700
PFTrDS	Perfluorotridecanesulfonic acid	791563–89-8	$C_{13}HF_{27}O_3S$	750
FOSA	Perfluorooctanesulfonamide	754–91-6	$C_8H_2F_{17}NO_2S$	499
DONA	4,8-Dioxa-3*H*-perfluorononanoic acid	919005–14-4	$C_7H_2F_{12}O_4$	378
HFPO-DA	Hexafluoropropylene oxide dimer acid	13252–13-6	$C_6HF_{11}O_3$	330
9Cl-PF3ONS	9-Chlorohexadecafluoro-3-oxanonane-1-sulfonic acid	756426–58-1	$C_8HClF_{16}O_4S$	532
11Cl-PF3OUdS	11-Chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	763051–92-9	$C_{10}HClF_{20}O_4S$	632
4:2FTS	1*H*, 1*H*, 2*H*, 2*H*-Perfluorohexane sulfonic acid	757124–72-4	$C_6H_5F_9O_3S$	328
6:2FTS	1*H*, 1*H*, 2*H*, 2*H*-Perfluorooctane sulfonic acid	27619–97-2	$C_8H_5F_{13}O_3S$	428
8:2FTS	1*H*, 1*H*, 2*H*, 2*H*-Perfluorodecane sulfonic acid	39108–34-4	$C_{10}H_5F_{17}O_3S$	528
10:2 FTS	1*H*, 1*H*, 2*H*, 2*H*-Perfluorododecane sulfonic acid	120226–60-0	$C_{12}H_5F_{21}O_3S$	628
Internal standard/surrogates				
M3PFBA	Perfluoro-n-[2,3,4-$^{13}C_3$]butanoic acid			217
M3PFPeA	Perfluoro-n-[3,4,5-$^{13}C_3$]pentanoic acid			267
M5PFHxA	Perfluoro-n-(1,2,3,4,6-$^{13}C_5$)hexanoic acid			316
M8PFOA	Perfluoro-n-[$^{13}C_8$]octanoic acid			422
MPFUdA	Perfluoro-n-(1,2-$^{13}C_2$)undecanoic acid			566
MPFDoA	Perfluoro-n-(1,2-$^{13}C_2$)dodecanoic acid			616
M2PFTeDA	Perfluoro-n-(1,2-$^{13}C_2$)tetradecanoic acid			716
M3PFBS	Sodium perfluoro-1-[2,3,4-$^{13}C_3$]butanesulfonate			303
M3PFHxS	Sodium perfluoro-1-(1,2,3-$^{13}C_3$)hexanesulfonate			404
M8PFOS	Sodium perfluoro-[$^{13}C_8$]octanesulfonate			508
M3HFPO-DA	2,3,3,3-Tetrafluoro-2-(1,1,2,2,3,3,3-heptafluoropropoxy)-$^{13}C_3$-propanoic acid			333
M8FOSA	Perfluoro-1-($^{13}C_8$)octanesulfonamide			507
$^{13}C_2$, D_4 4:2 FTS	1H, 1H, 2H, 2H-Perfluorohexane sulfonic acid (4:2 FTS) ($^{13}C_2$,99%; D_4,98%)			356
$^{13}C_2$, D_4 6:2 FTS	1H, 1H, 2H , 2H-Perfluorooctane sulfonic acid (4:2 FTS) ($^{13}C_2$,99%; D_4,98%)			456
$^{13}C_2$, D_4 8:2 FTS	1H, 1H , 2H , 2H-Perfluorodecane sulfonic acid (4:2 FTS) ($^{13}C_2$,99%; D_4,98%)			556
$^{13}C_2$, D_4 10:2 FTS	1H , 1H , 2H , 2H-Perfluorododecane sulfonic acid (4:2 FTS)($^{13}C_2$,99%; D_4,98%)			656
d_5-N-EtFOSAA	N-ethyl-d_5-perfluoro-1-octanesulfonamidoacetic acid			590

Add 15 mL water, 10 mL acetonitrile, and 150 μL formic acid to the 50 mL PP conical centrifuge tube and shake it vigorously for 1 min. Follow QuEChERS steps *(c)* and *(d)* as mentioned in *Sample preparation for food*.

(c) Filter 5 mL of supernatant with a 0.2 μm nylon syringe filter and transfer to a 15 mL conical centrifuge tube. Concentrate to about 1 mL using nitrogen in a 60°C water bath. Add 11 mL water in preparation for SPE analysis.

(3) *Concentrated method for specific matrixes (produce, milk, baby food).*—

(a) Add 5 g of food sample to a 50 mL PP centrifuge tube and add 5 μL 200 ng/mL isotopically labeled SS for a final concentration of 1 ng/mL in the final extract. Add 5 mL water, 10 mL acetonitrile, and 150 μL formic acid to the 50 mL PP conical centrifuge tube and shake it vigorously for 1 min. Follow QuEChERS steps *(c)* and *(d)* as mentioned in *Sample preparation for food*. Filter 5 mL supernatant with a 0.2 μm nylon syringe filter and transfer to a 15 mL conical centrifuge tube. Concentrate the 5 mL supernatant to 1 mL and add 11 mL water for SPE cleanup.

Table 3. Calibration curve preparation

Final concn, ng/mL	Native stock solution concn, ng/mL	Volume of stock solution to add, mL	Volume of 200 ng/mL surrogate stock solution, mL	Methanol, mL	Volume of 200 ng/mL IS stock solution, mL
0.01	10	0.01	0.05	9.94	0.05
0.05	10	0.05	0.05	9.9	0.05
0.1	10	0.1	0.05	9.85	0.05
0.5	100	0.05	0.05	9.9	0.05
1	100	0.1	0.05	9.85	0.05
5	1000	0.05	0.05	9.9	0.05
10	1000	0.1	0.05	9.85	0.05
25	1000	0.25	0.05	9.7	0.05

Table 4. Target matrixes used

Matrix category	Representative used	Sample size used
Produce	Kale, lettuce and apple juice	5 g and 5 mL
Coffee	Ground coffee (brewed before analysis)	5 mL
Milk (liquid)	Milk	5 mL
Dairy powders and plant-based protein powders	Whey protein powder (dairy based for adults)	1 g
Eggs	Boiled eggs	5 g
Seafood (crustaceans and molluss)	Shrimp and clams	5 g
Fish meat and meat of terrestrial animals	Salmon and cooked chicken breast	5 g
Edible offal of terrestrial animals	Beef liver (uncooked)	5 g
Fish oil	Fish oil	5 g
Foods for infants and young children (baby food)	Carrot, ginger, chickpea baby food puree	5 g
Pet food and animal feed	Corn snaplage	1 g

(4) *SPE clean-up using weak anion exchange column.*—
   (a) Condition a Strata™-XL-AW 100 μm column (200 mg/6 mL) with 6 mL 0.3% (w/v) ammonium hydroxide in methanol.
   (b) Add 5 mL LC/MS Optima water to equilibrate column. Add sample (the 12 mL diluted sample extract) to the column and let pass through the column at a flow rate of 1–2 drops/s.
   (c) Add 5 mL LC/MS Optima water to wash column. Dry for 1 min.
   (d) Add 4 mL 0.3% (w/v) ammonium hydroxide in methanol to the empty sample tube to rinse the sides and then add to the SPE cartridge to elute analytes into a clean 14 mL round-bottom PP tube.
   (e) Concentrate to below 0.5 mL using nitrogen in a 60°C water bath. Extracts should not be blown to dryness as certain analytes will have reduced recoveries.
   (f) Bring volume to 1 mL with methanol and add 5 μL 200 ng/mL d_5-N-EtFOSAA to the tube.
   (g) Transfer about 100 μL to a Thomson nano filter vial with 0.2 μm nylon filter and a nano filter screw cap (Thomson Instrument Co., Carlsbad, CA, USA) to run using LC–MS/MS.
   (h) For samples analyzed using the *Concentrated method for specific matrices*, after SPE elution, steps 4(*a*)–(*d*), the eluent is blow down using nitrogen to below 100 μL (do not allow to go dry). Reconstitute to 0.5 mL in methanol. Add 2.5 μL 200 ng/mL d_5-N-EtFOSAA to the tube. Transfer an aliquot to a Microsolv 300 μL PP autosampler vial with a screw-cap to run using LC–MS/MS.

(5) *PFOS cleanup using ENVI-Carb SPE tubes (for samples containing both PFOS and TCDCA).*—
   (a) Condition an ENVI-Carb cartridge 250 mg/6 mL (SupelClean ENVI-Carb, Millipore Sigma) with 4 mL methanol.
   (b) Elute 3 mL of the QuEChERS extract (acetonitrile) through the cartridge.
   (c) Concentrate to near dryness and reconstitute with 3 mL methanol and add internal standard (15 μL 0.2 ng/μL d_5-N-EtFOSAA, totaling 3 ng) and vortex mix.
   (d) Transfer a portion into a filter vial for LC–MS/MS analysis for PFOS determination only. This cleanup step is only performed on QuEChERS extracts that contain both PFOS and TCDCA.

Determination by LC–MS/MS

Analyze using a Shimadzu Nexera X2 HPLC interfaced with a SCIEX 6500+ mass spectrometer equipped with an ESI source. The sample sequence includes a blank (methanol) injection, a calibration curve, blank (methanol) injection, and samples. A CCV standard of 1 ng/mL is run after every six samples to check for accuracy. Accuracy of the CCV sample should fall within 70–130% of the original value.

System Suitability

System suitability data met the acceptance criteria outlined in SMPR 2023.003 prior to the acceptance of the validation data.

(a) *Selectivity.*—Verify selectivity by running procedural blanks with every batch. Take measures to reduce background contamination during sample extractions and avoid the use of fluoropolymers. A delay column placed between the mobile phase mixer and sample injector traps any system-related interferences. In the case of unavoidable background contamination, subtract the procedural blank concentrations from samples. Matrix blanks are considered suitable if the concentration in the matrix is no more than 30% of the target LOQ level for the given analyte. In the

case where no suitable matrix blank can be found (e.g., PFOS in beef liver), the LOQ is calculated by processing seven "blank" samples through the method and multiplying the sample standard deviation of the PFOS concentration in seven samples by 10 to calculate the LOQ.

(b) *Cholic acid interference.*—Certain food matrixes such as eggs, meat, milk, and pet food may show the presence of bile acids taurodeoxycholic acid (TDCA), taurochenodeoxycholic acid (TCDCA), and tauroursodeoxycholic acid (TUDCA). Cholic acids can elute at the same retention time as PFOS isomers and share the same ion transition, 499→80, as PFOS. In such cases chromatographic separation of cholic acids from all PFOS isomers must be achieved or these interferences need to be removed during extraction. Currently, the chromatographic method separates TUDCA and TDCA from all branched and linear PFOS isomers, but TCDCA interferes with PFOS. Therefore, an additional cleanup step with ENVI-Carb cartridges is needed to remove these cholic acids. A confirmatory PFOS MRM transition, 499→99, is also monitored which is interference-free but is less sensitive compared to the 499→80 ion transition. Additionally, the 499→124 MRM transition for all three cholic acids should be monitored to confirm the presence and subsequent removal of cholic acids from samples.

(c) *PFAS analytes with a single ion transition.*—For PFBA and PFPeA where only one ion transition is available, HRMS quantification is required to confirm detection and calculate the concentration of PFBA and PFPeA using accurate mass. Use the Shimadzu Nexera LC coupled to a Thermo Fisher Q-Exactive Orbitrap mass spectrometer with the same LC conditions as mentioned in *Appartus* section *(r)* of this method and mass spectrometer conditions as mentioned in *Apparatus* section *(t)*. Determination of the exact mass of PFBA (*m/z* 212.9792) and PFPeA (*m/z* 262.9760) with a ± 5 ppm mass accuracy tolerance and calculation of concentration is performed as described in *Apparatus* section *(t)* of this method using Thermo Scientific Xcalibur Quan Browser software.

(d) *Branched and linear isomers.*—Report PFHxS, PFOA, and PFOS concentrations as total PFHxS, total PFOA, and total PFOS, which is the sum of linear and branched isomers. Use MultiQuant (MQ4) peak integration parameters to integrate both branched and linear isomers; however, at lower concentrations manual integration may also be required. During integration of standards and samples, all peaks present are integrated as one peak and reported as the total PFOS, PFOA, or PFHxS concentration.

Table 5. Analytes with calibration curve fit and surrogates used as the internal standard

Surrogates with a mean response factor curve type, weighting none	
Surrogate standard	Internal standard
M3PFBA	d_5-N-EtFOSAA
M3PFPeA	d_5-N-EtFOSAA
M5PFHxA	d_5-N-EtFOSAA
M8PFOA	d_5-N-EtFOSAA
MPFUdA	d_5-N-EtFOSAA
MPFDoA	d_5-N-EtFOSAA
M2PFTeDA	d_5-N-EtFOSAA
M3PFBS	d_5-N-EtFOSAA
M3PFHxS	d_5-N-EtFOSAA
M8PFOS	d_5-N-EtFOSAA
M3HFPO	d_5-N-EtFOSAA
M8FOSA	d_5-N-EtFOSAA
$^{13}C_2$, D_4 4:2 FTS	d_5-N-EtFOSAA
$^{13}C_2$, D_4 6:2 FTS	d_5-N-EtFOSAA
$^{13}C_2$, D_4 8:2 FTS	d_5-N-EtFOSAA
$^{13}C_2$, D_4 10:2 FTS	d_5-N-EtFOSAA
Natives with a linear curve type, weighting 1/x	
Surrogate standard	Internal standard
PFBA	M3PFBA
PFPeA	M3PFPeA
PFHxA	M5PFHxA
PFHpA	M5PFHxA
PFOA	M8PFOA
PFNA	M8PFOA
PFDA	M8PFOA
PFUdA	MPFUdA
PFDoA	MPFDoA
PFTrDA	MPFDoA
PFTeDA	M2PFTeDA
PFBS	M3PFBS
PFPeS	M3PFHxS
PFHxS	M3PFHxS
PFHpS	M3PFHxS
PFOS	M8PFOS
PFNS	M8PFOS
PFDS	M8PFOS
PFUdS	MPFUdA
PFDoS	MPFDoA
PFTrDS	M2PFTeDA
FOSA	M8FOSA
DONA	M8PFOA
HFPO-DA	M3HFPO
9Cl-PF3ONS	M3PFHxS
11Cl-PF3OUdS	M3PFHxS
4:2 FTS	$^{13}C_2$, D_4 4:2 FTS
6:2 FTS	$^{13}C_2$, D_4 6:2 FTS
8:2 FTS	$^{13}C_2$, D_4 8:2 FTS
10:2 FTS	$^{13}C_2$, D_4 10:2 FTS

Calculations

(a) Use the SCIEX OS software to prepare a linear standard curve where x is the concentration ratio (analyte/SS), and y is the instrument response ratio (analyte/SS) with 1/x weighting. Surrogates and their internal standard pairs are listed in Table 5 which are used to calculate absolute recoveries of the surrogate standards over the entire extraction method. Surrogates and their native analyte pairs are also listed in Table 5 with their curve fit. The calibration curve has surrogate and internal standard concentrations of 1 ng/mL. In SCIEX OS, equations can be used to generate the surrogate recovery calculation instead of generating individual curves for each surrogate. Peak integration of all analytes is reviewed, and compounds with branched and linear isomers may require manual integration.

(b) Example calculations for concentration measured on LC–MS/MS (using the calibration curve) to concentration in food for the different method modifications:

(1) *For food.*—Multiply the calculated concentrations from the instrument (ppb) by a dilution factor which considers the volume of extraction solvent (10 mL) and the mass of food (5 g) to produce concentration in food (in ppb). For example, if the calculated concentration from the instrument is 1 ng/mL with a 10 mL extraction volume and 5 g food, the concentration in food is 2 ng/g:

$$\frac{1\ ng}{mL} * \frac{10\ mL}{5\ grams} = 2\ ng/g$$

(2) *For feed.*—For feed samples, use a 10 mL extraction volume (where 5 mL extract is concentrated to 1 mL), and 1 g food. If the calculated concentration from the instrument is 1 ng/mL, then the concentration in food is 2 ng/g from the calculation shown below:

$$\frac{1\ ng}{mL} * \frac{10\ mL}{1\ grams} * \frac{1\ mL}{5\ mL} = 2\ ng/g$$

(3) *For the concentrated method.*—These samples have a 10 mL extraction volume (where 5 mL extract is concentrated to 1 mL) and 5 g food. In addition, the final volume is concentrated from 1 mL to 0.5 mL. Calculate concentrations from the instrument (ppb) by dividing by 5 to get the concentration in food in ppb. If the calculated concentration from the instrument is 1 ng/mL, the final concentration in food is 0.2 ng/g as shown in the calculation below:

$$\frac{1\ ng}{mL} * \frac{0.5mL}{1mL} * \frac{10\ mL}{5\ grams} * \frac{1\ mL}{5\ mL} = 0.2\ ng/g$$

Results and Discussion

(a) *Method validation.*—Method validation was performed at the target LOQ and at two other concentration levels in the range of 2–100× of the target LOQ level for each matrix category as specified in SMPR 2023.003. Based on the data obtained, two additional concentration levels (0.05 and 0.1 µg/kg) below the target LOQ level were also performed except for matrixes such as produce and milk where the required LOQ is ≤0.01 µg/kg. Method performance was evaluated by confirming all system suitability requirements were met. All samples were evaluated for selectivity, linearity, LOQ, and recovery not only for selected PFAS analytes—PFOS, PFOA, PFNA, PFHxS mentioned in SMPR 2023.003—but also other PFAS analytes in

Table 6. Spiking levels for PFAS analytes (except for PFBA and PFPeA)

Matrix used	Method LOQ, µg/kg	AOAC target LOQ, µg/kg	Spike 1, µg/kg	Spike 2, µg/kg	Spike 3, µg/kg	Spike 4, µg/kg
Ground coffee (brewed before analysis)	0.05	0.3	3	9	30	—
Whey protein powder (dairy based for adults)	0.08	0.08	0.8	2.4	3	—
Shrimp	0.05	0.3	0.1	3	9	—
Cooked chicken breast	0.05	0.1	1	3	10	—
Beef liver (uncooked)[a]	0.05	0.4	1	4	—	—
Fish oil	0.05	0.5	0.1	5	15	—
Corn snaplage	0.05	0.5	0.1	5	15	—
Boiled eggs	0.05	0.3	0.1	1[b]	5[b]	15[b]
Clams	0.05	0.3	0.1	1[b]	5[b]	15[b]
Salmon (uncooked)	0.05	0.1	1[b]	5[b]	15[b]	—[c]
Concentrated method						
Kale and apple juice	0.01	0.01	0.1	0.3	—	—
Milk, lettuce	0.01	0.01	0.02	0.1	0.5	—
Carrot, ginger, chickpea baby food puree	0.01	0.01	0.1	0.3	—	—

[a] LOQ for PFOS, PFOA, and PFHxA was calculated based on seven matrix samples.
[b] SLV data from reference (8).
[c] — = Dash indicates matrix/spike combination not tested

Table 7. Spiking levels for PFBA and PFPeA

Matrix	Target LOQ, µg/kg	Spike 1, µg/kg	Spike 2, µg/kg	Spike 3, µg/kg
Fruit juice (apple) concentrated	1	5	10	—[b]
Kale concentrated	1	5	10	—
Baby food concentrated	1	0.1	0.3	10
Milk concentrated	1	0.1	0.5	10
Beef liver	4	0.4	1	—
Liquid coffee	3	0.3	9	30
Shrimp	3	0.3	1	9
Chicken	1	3	10	—
Fish oil	5	0.5	1	15
Dairy based protein powder	1	0.8	2.4	3
Snaplage sample	5	3	15	—
Lettuce concentrated	1	0.5	10	—
Eggs[a]	1	5	15	—
Clams[a]	1	5	15	—
Salmon[a]	1	5	15	—

[a] SLV data from reference (8).
[b] — = Dash indicates matrix/spike combination not tested.

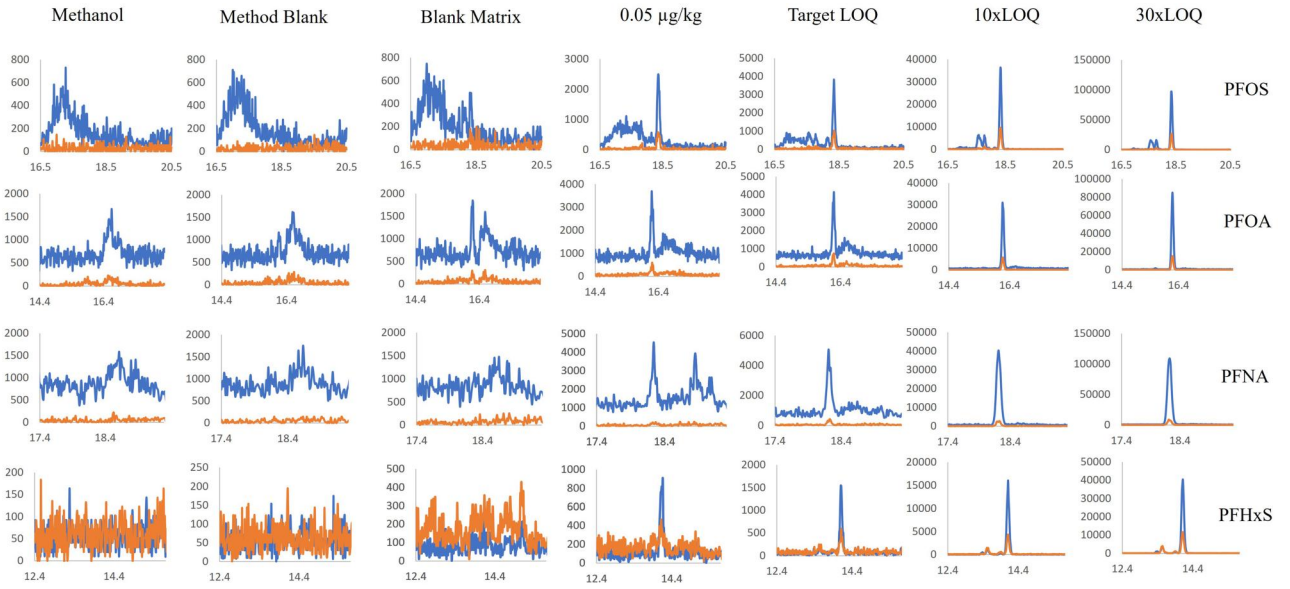


Figure 2. Extracted ion chromatograms of PFOA, PFOS, PFNA, and PFHxS in chicken. The top trace represents the quantifying ion transition, and the lower trace represents the qualifying ion transition. Spike level for chicken at method LOQ is 0.05 µg/kg; target LOQ is 0.1 µg/kg; 10× LOQ is 1 µg/kg and 30× LOQ is 3 µg/kg.

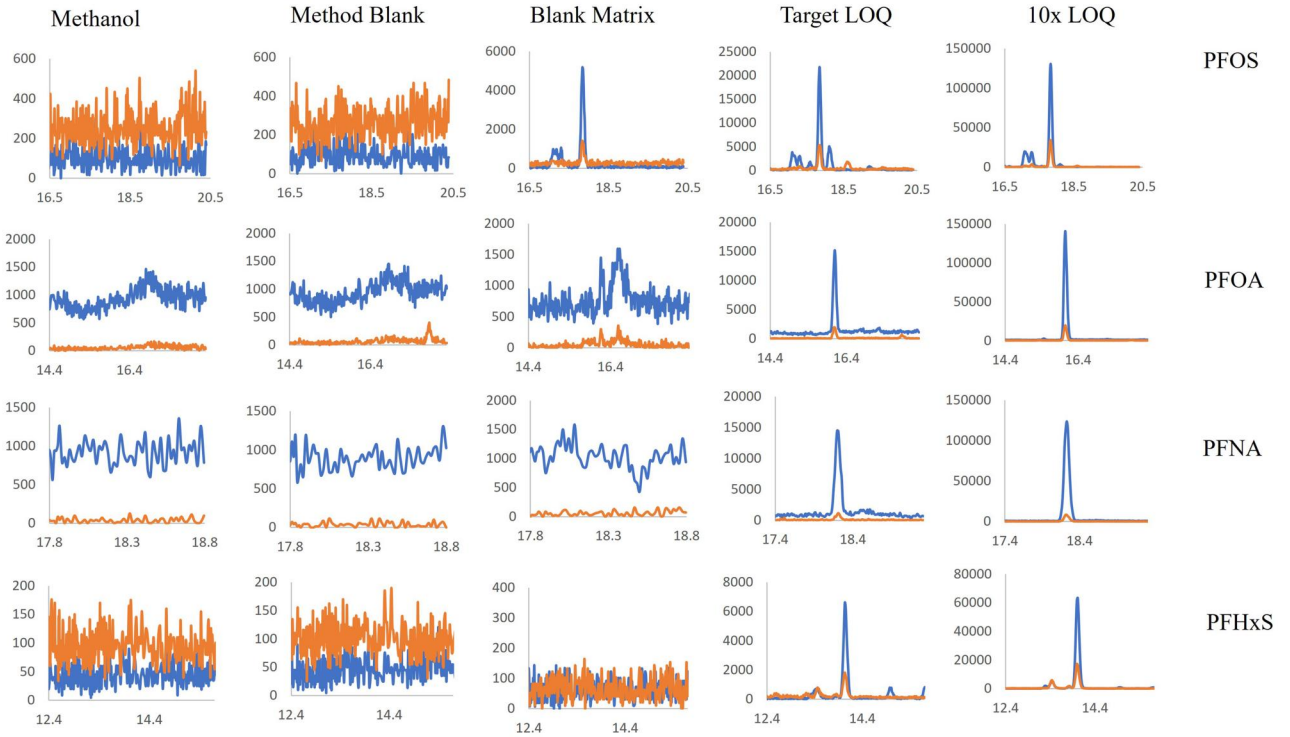


Figure 3. Extracted ion chromatograms of PFOS, PFOA, PFNA, and PFHxS in beef liver. The top trace represents the quantifying ion transition, and the lower trace represents the qualifying ion transition. Spike level for beef liver at target LOQ is 0.4 µg/kg and 10× LOQ is 4 µg/kg.

the FDA method. Method performance was evaluated by spiking a suitable matrix blank in triplicate for each matrix category mentioned in Table 4. Spiking concentrations for each analyte are given in Table 6 except for PFBA and PFPeA. Spiking concentrations for PFBA and PFPeA are given in Table 7. Extracted ion chromatograms for PFOS, PFOA, PFNA, and PFHxS are shown in Figures 2–5 for chicken, beef liver, shrimp, and eggs.

(1) *LOQ.*—LOQ for EU-regulated matrixes and for other matrixes are summarized in Tables 8 and 9, respectively. LOQs for PFOS, PFOA, PFNA, and PFHxS were 0.05 µg/kg in eggs, shrimp, salmon, chicken, and clams. LOQ calculation for beef liver matrix will be discussed separately in the following section due to unavoidable background contamination in the blank beef liver matrix. For other PFAS analytes LOQs ranged from 0.05 to 0.1 µg/kg and

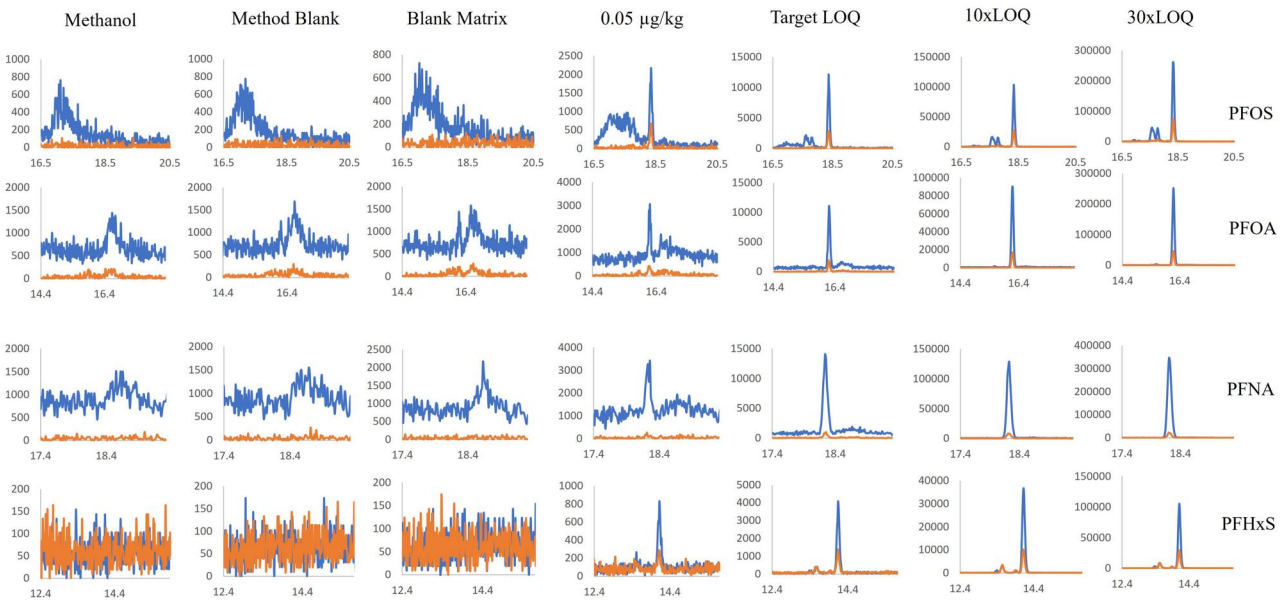


Figure 4. Extracted ion chromatograms of PFOS, PFOA, PFNA, and PFHxS in shrimp The top trace represents the quantifying ion transition, and the lower trace represents the qualifying ion transition. Spike level for shrimp at method LOQ is 0.05 µg/kg; target LOQ is 0.3 µg/kg; 10× LOQ is 3 µg/kg and 30× LOQ is 9 µg/kg.

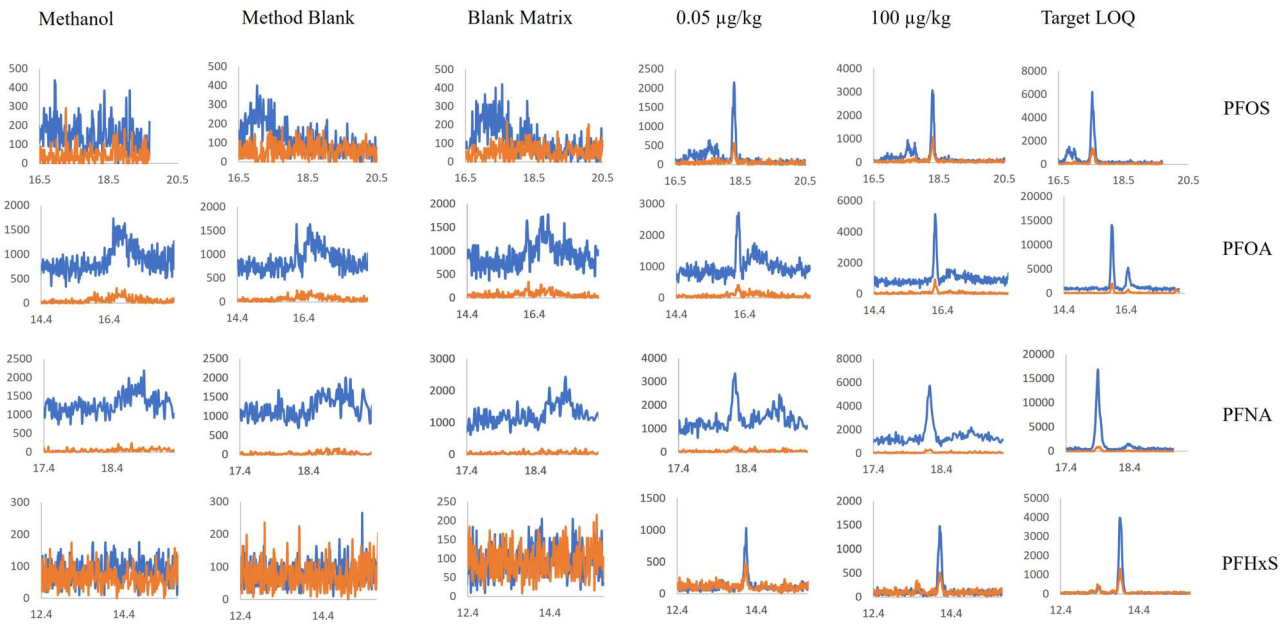


Figure 5. Extracted ion chromatograms of PFOS, PFOA, PFNA, and PFHxS in eggs. The top trace represents the quantifying ion transition, and the lower trace represents the qualifying ion transition. Spike levels for eggs are at 0.05 µg/kg, at 0.1 µg/kg, and at target LOQ, 0.3 µg/kg.

are summarized in Tables S1 and S2 in the supplemental information. LOQs were established and validated only when the following three positive identification criteria were met simultaneously.

(*a*) The retention time of each analyte and internal standard matches with the average retention time of the calibration points in the same sequence with a tolerance of 1%. Data for the retention time validation parameter is provided in Table S5 in the supplemental information.

(*b*) The S/N of the confirmation ion transition for the lowest calibration point was ≥3. The S/N integration threshold was set to 3 in the integration parameters of Sciex OS software to integrate peaks that have S/N ≥3.

(*c*) The ion ratios of the diagnostic ions corresponded to those in the calibration points of the same sequence with ±30% relative tolerance.

All the above mentioned system suitability criteria were systematically applied to the target LOQ and

Table 8. LOQ for PFOS, PFOA, PFNA, and PFHxS in EU-regulated matrixes compared to target LOQs

Matrix	Analyte	AOAC target LOQ, μg/kg	LOQ, μg/kg
Eggs	PFOS	0.30	0.05
	PFOA	0.30	0.05
	PFNA	0.30	0.05
	PFHxS	0.30	0.05
Shrimp	PFOS	0.30	0.05
	PFOA	0.30	0.05
	PFNA	0.30	0.05
	PFHxS	0.30	0.05
Clams	PFOS	0.30	0.05
	PFOA	0.30	0.05
	PFNA	0.30	0.05
	PFHxS	0.30	0.05
Salmon	PFOS	0.10	0.05
	PFOA	0.10	0.05
	PFNA	0.10	0.05
	PFHxS	0.10	0.05
Chicken	PFOS	0.10	0.05
	PFOA	0.10	0.05
	PFNA	0.10	0.05
	PFHxS	0.10	0.05
Beef liver	PFOS	0.40	0.06[a]
	PFOA	0.40	0.08[a]
	PFNA	0.40	0.05
	PFHxS	0.40	0.05

[a] LOQ calculated based on seven blank matrix samples.

Table 9. LOQ for PFOS, PFOA, PFNA, and PFHxS in other matrixes compared to target LOQs

Matrix	Analyte	AOAC target LOQ, μg/kg	LOQ, μg/kg
Apple juice	PFOS	0.01	0.01
	PFOA	0.01	0.01
	PFNA	0.01	0.01
	PFHxS	0.01	0.01
Kale	PFOS	0.01	0.01
	PFOA	0.01	0.01
	PFNA	0.01	0.01
	PFHxS	0.01	0.01
Lettuce	PFOS	0.01	0.01
	PFOA	0.01	0.01
	PFNA	0.01	0.01
	PFHxS	0.01	0.01
Baby food	PFOS	0.01	0.01
	PFOA	0.01	0.01
	PFNA	0.01	0.01
	PFHxS	0.01	0.01
Milk	PFOS	0.01	0.01
	PFOA	0.01	0.01
	PFNA	0.01	0.01
	PFHxS	0.01	0.01
Ground coffee	PFOS	0.3	0.05
	PFOA	0.3	0.05
	PFNA	0.3	0.05
	PFHxS	0.3	0.3
Whey protein powder	PFOS	0.08	0.08
	PFOA	0.08	0.08
	PFNA	0.08	0.08
	PFHxS	0.08	0.08
Fish oil	PFOS	0.5	0.05
	PFOA	0.5	0.05
	PFNA	0.5	0.05
	PFHxS	0.5	0.05
Corn snaplage	PFOS	0.5	0.05
	PFOA	0.5	0.05
	PFNA	0.5	0.05
	PFHxS	0.5	0.05

Table 10. Summary of compounds that had unsatisfactory data (see also Tables S1 and S2 in the supplemental information)

Matrix	Compounds with unsatisfactory data
Eggs	No qualifying ion peak observed for PFPeS, 9Cl-PF3ONS, and FOSA at 0.05 μg/kg concentration and no qualifying ion peak for FOSA at 0.1 μg/kg. All pass at target LOQ of 0.3 μg/kg.
Shrimp	11Cl-PF3OUdS and FOSA did not show a qualifying ion peak at target LOQ concentration of 0.3 μg/kg. PFDoA, 9Cl-PF3ONS, and FOSA did not show a qualifying ion peak at 0.05 μg/kg. Interference was observed for PFDoA at 0.05 and 0.1 μg/kg levels. However, all these analytes passed at 0.1 μg/kg, which is below the 0.3 μg/kg target LOQ.
Clams	PFDoA shows an interference peak partially eluting with the quantifying peak and not in the qualifier ion transition but ion ratios pass for all spikes.
Salmon	Qualifying ion peak for PFPeS shows matrix interference and ion ratios do not pass for 0.05 μg/kg. No qualifying ion peaks observed for FOSA and HFPO-DA at 0.05 μg/kg level. All pass at the targeted LOQ of 0.1 μg/kg.
Brewed coffee	PFHxS did not meet requirements at 0.05 μg/kg level but met target LOQ of 0.3 μg/kg No qualifying ion peak detected at 0.05 μg/kg level for PFPeS, HFPO-DA, 9Cl-PF3ONS, 11Cl-PF3OUdS, and FOSA, but all met target LOQ of 0.3 μg/kg except 11Cl-PF3OUdS and FOSA, which pass at 3 μg/kg.
Corn snaplage	No LOQ determination could be made for PFTeDA and 4:2 FTS in snaplage due to matrix interference issues. PFHpA and PFBS do not meet requirements at 0.05 μg/kg level but pass at 0.1 μg/kg below the target LOQ of 0.5 μg/kg. Qualifying ion peaks were not observed for 11Cl-PF3OUdS at target LOQ level and for FOSA at 0.05 and 0.1 μg/kg, but 11Cl-PF3OUdS passes at 0.1 μg/kg and FOSA passes at target LOQ of 0.5 μg/kg. At 0.05 and 0.1 μg/kg levels, HFPO-DA shows an interference co-eluting with the quantifier ion peak, but the ion ratio still passes. Accurate analysis of 4:2 FTS in corn snaplage requires the use of HRMS.
Protein powder	PFHxA does not meet requirements at target LOQ level. PFOA observed in blank matrix.
Kale	PFHxS interference with qualifying ion is observed in target LOQ, 0.1 μg/kg, and 0.3 μg/kg levels, but ion ratios still pass at these concentrations.

other concentration levels. Data for salmon, clams, and eggs at 0.15, 1, 5, and 15 μg/kg spiking concentrations were previously validated and published in a peer-reviewed journal article (8). A summary of the compounds that did not meet all requirements is provided in Table 10.

For matrixes such as produce, baby food, and milk, the required LOQs set by SMPR 2023.003 are ≤0.01 μg/kg. Calculated LOQs using the FDA method mentioned in sample preparation section *Sample preparation for food (matrixes such as chicken, shrimp, salmon, clams, beef liver, pet food, brewed coffee)* were above the required LOQ. To decrease the LOQ additional concentration steps were introduced as mentioned in sample preparation section *Concentrated method for specific matrixes (produce, milk, baby food)*.

(2) *Matrix blank*.—To assess method selectivity, suitable matrix blanks were chosen that contained no more than 30% of the target LOQ for the given analyte. Additionally, procedural blanks were run to determine any other

Table 11. Recoveries and repeatability for PFOS, PFOA, PFNA, and PFHxS

		Average recovery, %						Repeatability (RSD), %					
Matrix	**Analyte**	**Method LOQ**	**Target LOQ**	**Spike 1**	**Spike 2**	**Spike 3**	**Spike 4**	**Method LOQ**	**Target LOQ**	**Spike 1**	**Spike 2**	**Spike 3**	**Spike 4**
Chicken	PFOS	97	105	101	108	95	—	6	8	5	4	1	—
	PFOA	83	97	110	97	98	—	3	3	1	4	3	—
	PFNA	115	97	104	94	100	—	0	3	3	2	3	—
	PFHxS	93	98	99	95	93	—	5	12	3	2	1	—
Shrimp	PFOS	83	98	110	99	99	—	0	2	10	1	1	—
	PFOA	112	100	113	100	99	—	0	6	5	4	1	—
	PFNA	104	94	104	97	102	—	0	8	0	1	2	—
	PFHxS	104	102	91	96	97	—	0	2	9	1	1	—
Beef liver	PFOS	[b]	86	95	93	—	—	[b]	5	2	5	—	—
	PFOA	[b]	99	101	100	—	—	[b]	2	3	1	—	—
	PFNA	109	93	101	93	—	—	9	4	1	5	—	—
	PFHxS	92	107	101	94	—	—	4	3	1	3	—	—
Ground coffee	PFOS	71	106	94	97	96	—	3	2	5	1	3	—
	PFOA	113	103	97	98	93	—	7	2	2	1	1	—
	PFNA	95	97	94	97	91	—	2	2	1	1	1	—
	PFHxS	96	94	93	97	94	—	8	5	2	0	0	—
Whey protein powder	PFOS	[c]	108	104	95	91	—	[c]	14	4	18	11	—
	PFOA	[c]	109	105	93	91	—	[c]	14	5	16	11	—
	PFNA	[c]	92	98	94	91	—	[c]	23	3	15	13	—
	PFHxS	[c]	98	99	91	90	—	[c]	15	5	15	10	—
Fish oil	PFOS	81	104	100	103	103	—	8	4	0	2	0	—
	PFOA	121	116	103	108	107	—	4	3	0	3	2	—
	PFNA	101	107	107	107	106	—	0	2	0	3	1	—
	PFHxS	80	105	94	101	101	—	9	2	0	1	1	—
Corn snaplage	PFOS	93	112	103	111	103	—	7	5	9	1	1	—
	PFOA	72	131	73	113	106	—	11	3	10	2	3	—
	PFNA	111	111	99	111	105	—	2	3	18	1	2	—
	PFHxS	108	107	83	109	105	—	7	3	6	1	2	—
Salmon	PFOS	101	113	85[a]	86[a]	94[a]	—	8	7	1[a]	8[a]	7[a]	—
	PFOA	101	111	88[a]	87[a]	101[a]	—	5	3	3[a]	8[a]	3[a]	—
	PFNA	84	105	91[a]	92[a]	100[a]	—	5	0	3[a]	9[a]	3[a]	—
	PFHxS	97	93	93[a]	90[a]	98[a]	—	10	2	3[a]	8[a]	2[a]	—
Eggs	PFOS	92	96	110	97[a]	94[a]	90[a]	0	0	3	2[a]	2[a]	3[a]
	PFOA	91	98	102	94[a]	92[a]	100[a]	9	3	3	2[a]	1[a]	4[a]
	PFNA	113	96	97	96[a]	91[a]	95[a]	0	4	0	4[a]	2[a]	8[a]
	PFHxS	99	95	95	94[a]	91[a]	99[a]	8	4	6	3[a]	3[a]	3[a]
Clams	PFOS	88	105	90	97[a]	96[a]	89[a]	5	0	9	2[a]	3[a]	4[a]
	PFOA	120	96	109	97[a]	97[a]	96[a]	3	1	4	2[a]	1[a]	1[a]
	PFNA	119	96	107	99[a]	99[a]	91[a]	14	5	4	1[a]	1[a]	3[a]
	PFHxS	105	96	88	94[a]	92[a]	95[a]	8	6	4	3[a]	2[a]	1[a]
Concentrated method													
Baby food	PFOS	[c]	123	113	105	—	—	[c]	8	6	13	—	—
	PFOA	[c]	116	111	105	—	—	[c]	3	2	13	—	—
	PFNA	[c]	96	112	105	—	—	[c]	4	1	12	—	—
	PFHxS	[c]	92	112	105	—	—	[c]	9	2	13	—	—
Apple juice	PFOS	[c]	75	118	116	—	—	[c]	1	3	2	—	—
	PFOA	[c]	103	113	116	—	—	[c]	3	3	2	—	—
	PFNA	[c]	103	101	106	—	—	[c]	1	1	4	—	—
	PFHxS	[c]	126	114	115	—	—	[c]	3	5	3	—	—
Kale	PFOS	[c]	112	114	113	—	—	[c]	12	10	1	—	—
	PFOA	[c]	108	111	113	—	—	[c]	6	0	2	—	—
	PFNA	[c]	117	116	122	—	—	[c]	1	1	1	—	—
	PFHxS	[c]	100	113	115	—	—	[c]	2	1	1	—	—
Milk	PFOS	[c]	108	110	112	121	—	[c]	6	4	1	2	—
	PFOA	[c]	104	100	104	107	—	[c]	11	6	2	1	—
	PFNA	[c]	89	103	103	107	—	[c]	3	6	3	2	—
	PFHxS	[c]	109	113	106	106	—	[c]	11	3	1	2	—
Lettuce	PFOS	[c]	103	110	112	106	—	[c]	0	4	1	4	—
	PFOA	[c]	103	100	108	111	—	[c]	15	6	3	4	—
	PFNA	[c]	89	103	108	114	—	[c]	9	6	3	3	—
	PFHxS	[c]	100	113	104	107	—	[c]	5	5	1	2	—

[a] Spikes 1, 2, and 3 for salmon and Spikes 2, 3, and 4 for eggs and clams are from SLV data, reference (8); concentration of spike levels are in Table 6.
[b] LOQ calculated with seven blank matrix samples.
[c] Same as target LOQ.

Table 12. LOQs for PFBA and PFPeA compared to target LOQs

Matrix	Analyte	Target LOQ, μg/kg	LOQ, μg/kg
Eggs	PFBA	1	1
	PFPeA	1	1
Shrimp	PFBA	3	0.3
	PFPeA	3	0.3
Clams	PFBA	1	1
	PFPeA	1	1
Salmon	PFBA	1	1
	PFPeA	1	1
Chicken	PFBA	1	1
	PFPeA	1	1
Beef liver	PFBA	4	0.4
	PFPeA	4	0.4
Apple juice	PFBA	1	1
	PFPeA	1	1
Kale	PFBA	1	1
	PFPeA	1	1
Lettuce	PFBA	1	0.5
	PFPeA	1	0.5
Baby food	PFBA	1	0.1
	PFPeA	1	0.1
Milk	PFBA	1	0.1
	PFPeA	1	0.1
Ground coffee	PFBA	3	0.3
	PFPeA	3	0.3
Whey protein powder	PFBA	1	0.8
	PFPeA	1	1
Fish oil	PFBA	5	0.5
	PFPeA	5	0.5
Corn snaplage	PFBA	5	3
	PFPeA	5	3

interferences caused by background contamination. Beef liver blank matrix showed >30% of the target LOQ of 0.05 μg/kg for PFOS, PFOA, and PFHxA. Therefore, the LOQ was determined by extracting seven blank matrix samples using the extraction protocol mentioned in sample preparation section *Sample preparation for food*. The LOQ calculation was performed using the following equation, where *s* is the sample standard deviation of seven blank matrix samples. Based on this, the calculated LOQs for PFOS, PFOA, and PFHxA in beef liver are 0.06 μg/kg, 0.08 μg/kg, and 0.06 μg/kg respectively.

$$LOQ = 10{*}s$$

For sample matrixes such as beef liver and eggs, where cholic acid interferences were present, additional cleanup was carried out using ENVI-Carb cartridges and MS/MS ion transition 499→124 was monitored to confirm removal of cholic acids from samples.

(b) *Recoveries and repeatability.*—As per SMPR 2023.003 acceptable recovery percentages for PFOS, PFOA, PFHxS, and PFNA should fall between 80 and 120% for EU-regulated matrixes and 65–135% for other matrixes. Repeatability (RSD_r) for these selected PFAS should be ≤20% for EU-regulated matrixes and ≤25% for other matrixes. For other PFAS analytes without commercially available matching isotopically labeled standards, recoveries within 40–140% and RSD_r ≤30% are acceptable.

Table 13. Recoveries and repeatability of PFBA and PFPeA using HRMS

		Average percent recovery, %				Repeatability (RSD), %			
Matrix	Analyte	Target LOQ	Spike 1	Spike 2	Spike 3	Target LOQ	Spike 1	Spike 2	Spike 3
Eggs[a]	PFBA	99	91	96	—	2	4	3	—
	PFPeA	89	89	95	—	3	3	1	—
Shrimp	PFBA	88	95	104	91	2	9	4	2
	PFPeA	98	104	102	98	3	6	2	2
Clams[a]	PFBA	93	87	91	—	1	1	2	—
	PFPeA	91	92	90	—	9	3	5	—
Salmon[a]	PFBA	83	88	93	—	2	12	4	—
	PFPeA	84	88	97	—	4	11	3	—
Chicken	PFBA	113	103	101	—	5	1	4	—
	PFPeA	100	98	103	—	2	1	2	—
Beef liver	PFBA	118	93	113	—	13	1	10	—
	PFPeA	100	91	96	—	5	4	4	—
Apple juice	PFBA	104	98	111	—	1	3	1	—
	PFPeA	104	99	110	—	1	2	4	—
Kale	PFBA	119	134	133	—	1	10	3	—
	PFPeA	131	103	116	—	4	12	1	—
Lettuce	PFBA	124	97	121	—	12	3	10	—
	PFPeA	117	120	126	—	9	5	6	—
Baby food	PFBA	103	133	102	—	54	17	18	—
	PFPeA	100	119	103	—	45	5	12	—
Milk	PFBA	118	119	117	120	1	5	2	3
	PFPeA	121	113	115	113	8	4	3	2
Ground coffee	PFBA	100	93	110	111	0	2	2	1
	PFPeA	91	95	90	98	2	5	3	2
Whey protein powder	PFBA	185[b]	69	118	105	108[b]	17	16	15
	PFPeA	110	13[b]	109	93	22	0	23	9
Fish oil	PFBA	90	99	103	94	2	9	7	1
	PFPeA	99	103	96	93	5	1	2	1
Corn snaplage	PFBA	120	110	112	—	12	5	6	—
	PFPeA	118	102	117	—	4	2	10	—

[a] Target LOQ, spikes 1 and 2 for eggs, clams, and salmon is from SLV data, reference (8).
[b] Did not meet SMPR 2023.03 requirements.

Recoveries and reproducibility data for PFOS, PFOA, PFNA, and PFHxS in EU-regulated matrixes (eggs, shrimp, clams, salmon, chicken, and beef liver) and other matrixes (apple juice, kale, lettuce, baby food, milk, ground coffee, whey protein powder, and corn snaplage) were within acceptable ranges as summarized in Table 11 but with a few exceptions as mentioned in Table 10. Percentage recoveries and repeatability for other PFAS analytes measured at different spiking levels are summarized in Tables S1 and S2 provided in the supplemental information. PFBA and PFPeA data will be discussed separately as their confirmation and quantification is based on HRMS.

Recovery and repeatability data for salmon, clams, and eggs were validated, and data were previously published in a peer-reviewed journal article (8). Data from this study are also included in Tables S1 and S2.

(c) *Reference materials*.—The method applicability was further demonstrated through participating in a proficiency testing program of spiked food matrixes from the National Measurement Institute (NMI), which included fish, spinach, and bovine liver, in July of 2023. Method accuracy data from these test materials for each PFAS analyte are summarized in Table S4 in the supplemental information. Results from each test material were compared against the assigned value as mentioned in Table S4. Proficiency tests final report "AQA 23-15 PFAS in biota and food" from a collaborative study between 22 laboratories is currently available on the Department of Industry, Science and Resources, Australian Government website in which our lab code is 17 (12).

(d) *HRMS confirmation of PFBA and PFPeA*.—HRMS confirmation and quantification of PFBA and PFPeA was performed using a Q-Exactive Orbitrap high-resolution mass spectrometer. LOQs for PFBA and PFPeA in EU-regulated and other matrixes compared to target LOQs set in SMPR 2023.003 are summarized in Table 12. The LOQ for PFBA and PFPeA in eggs, clams, salmon, chicken, and apple juice is 1 ppb. The LOQ for PFBA and PFPeA in shrimp is 0.3 μg/kg and 0.4 μg/kg in beef liver. The LOQs for PFBA and PFPeA for kale, lettuce, coffee, whey protein powder, fish oil, and corn snaplage ranged from 0.1 to 0.8 μg/kg.

Acceptable recoveries for PFBA and PFPeA should be between 65 and 135% with ≤25% repeatability. Recoveries and repeatability for PFBA and PFPeA at different spike levels are summarized in Table 13. All were within acceptable ranges with a few exceptions highlighted in Table 13.

Conclusions

The method described in this manuscript has been validated for the determination of 30 PFAS analytes using an isotope dilution LC–MS/MS method. The data presented in this paper were generated through a single-laboratory validation in produce, beverages, dairy products, eggs, seafood, meat products, and feed. The results show that the proposed analytical validation meets SMPR 2023.003.

CRediT Author Statement

Manjula Sunkara (Data curation [Lead], Formal analysis [Lead], Validation [Lead], Writing—original draft [Lead], Writing—review & editing [Lead]), Elsie Peprah (Formal analysis [Equal], Validation [Equal]), Wendy Young (Formal analysis [Equal], Validation [Equal]), Samanta Popol (Formal analysis [Equal], Validation [Equal]), Maria Moreno-Sanitago (Formal analysis [Equal], Validation [Equal]), Brian Ng (Formal analysis [Equal], Validation [Equal], Writing—original draft [Supporting], Writing—review & editing [Supporting]), Christine M. Fisher (Writing—original draft [Supporting], Writing—review & editing [Supporting]), Lowri DeJager (Project administration [Lead]), and Susan Genualdi (Conceptualization [Lead], Investigation [Lead], Project administration [Lead], Writing—review & editing [Supporting])

Conflict of Interest

All authors declare no conflict of interest.

Supplemental Information

Supplemental information is available on the *J. AOAC Int.* website.

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Journal of AOAC International, 2026, 109, 511–525
https://doi.org/10.1093/jaoacint/qsag007
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